

Open-Drain

This open-drain signalling ensures safe, collision-free communication:

- Any device can pull the line LOW.
- No device ever forces it HIGH.
- If two devices send data at once, the LOW signal always dominates, preventing conflict.

This design enables two powerful features:

Arbitration: Multiple masters can attempt communication without corrupting data.

Clock stretching: Slower devices can hold the clock LOW to pause communication until they're ready.

This is critical in a server motherboard, where various controllers (such as the CPU's power controller and BMC) may share the same I2C bus. Without open-drain logic, they could interfere with each other and destabilise system monitoring.

SCL line

This is how the SCL line works in I2C communication:

- **Master pulls SCL LOW:** The master actively pulls the SCL line LOW to

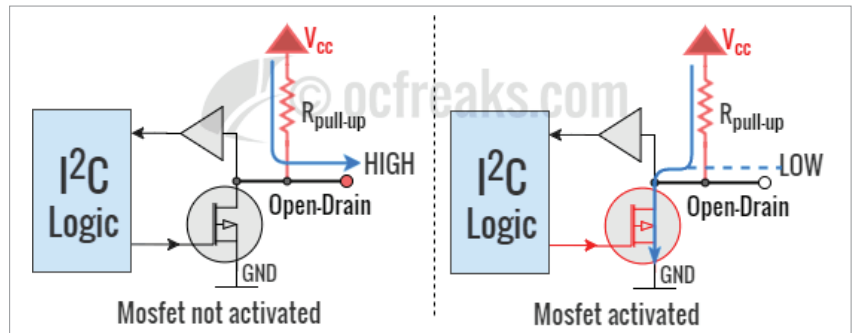


Figure 2: Open-drain in I2C (Image Source: <https://www.ocfreaks.com/i2c-tutorial/>)

begin a clock cycle.

- **While SCL is LOW, the transmitter sets the data (SDA):** The device transmitting the data (master or slave) drives the SDA line to either logic 0 or logic 1.
- Data is allowed to change only when the SCL line is LOW.
- **Master releases SCL (floating state):** The master stops pulling SCL LOW and releases the line.
- **Pull-up resistor pulls SCL HIGH:** Since no device is pulling the line LOW, the pull-up resistor drives SCL HIGH.
- **When SCL is HIGH, the receiver**

samples SDA: The receiving device reads the SDA line while SCL is HIGH. The data must remain stable during this period.

Steps of I2C data transmission

The I2C follows the steps given below to transmit data (as shown in Figure 3):

Start condition: The master device sends a start condition by pulling the SDA line low while the SCL line is high. This signals that a transmission is about to begin.

Addressing the slave: The master sends the 7-bit address of the slave device it wants to communicate with,

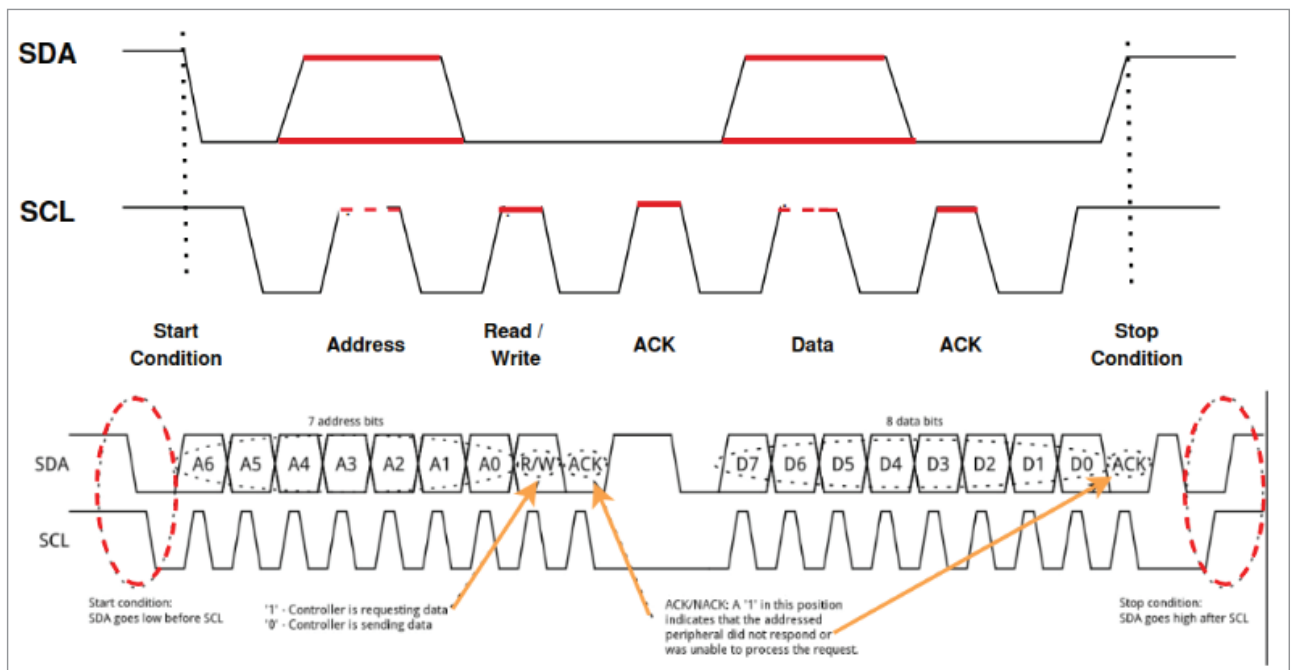


Figure 3: Steps in I2C communication